



NEWSBOT

INTELLIGENCE ENGINE

— 2.0 —

# NewsBot Intelligence Engine 2.0

TECHNICAL DOCUMENTATION

ITAI 2373 — NATURAL LANGUAGE PROCESSING

FINAL PROJECT | HOUSTON CITY COLLEGE



**LEROY BROWN**

Intelligent Systems Built to Scale

 May 2026



**2,000**

BBC ARTICLES  
PROCESSED



**97.6%**

CLASSIFICATION  
ACCURACY



**5+**

LANGUAGES  
SUPPORTED



**4**

ADVANCED  
MODULES



**7**

NL INTENTS  
SUPPORTED

# 1. Project Overview



NewsBot Intelligence Engine 2.0 is the final-project extension of the midterm NewsBot system, building on a proven 97.6%-accuracy classification pipeline to add four advanced NLP modules: topic modeling, language generation, multilingual intelligence, and a conversational query interface. The system processes 2,000 BBC news articles across five categories and exposes its capabilities through both a Gradio chat UI and a production-grade Next.js web dashboard.

## Scope & Objectives

-  **Extend:** Build directly on the midterm codebase — same dataset, same preprocessing, same best model (Logistic Regression, 97.6%). No regression in baseline capability.
-  **Topic Modeling:** Apply LDA with 10 topics across all five categories and surface per-category topic signatures using pyLDAvis for interactive exploration.
-  **Summarization:** Deploy DistilBART (sshleifer/distilbart-cnn-12-6) for CPU-feasible abstractive summarization with average ~60% compression ratio.
-  **Semantic Search:** Embed 500-article index with all-MiniLM-L6-v2 sentence transformers enabling cosine-similarity retrieval against natural language queries.
-  **Multilingual:** Integrate langdetect for language identification and deep-translator (Google Translate API) to enable classification of non-English news.
-  **Conversational:** Deploy a 7-intent NL query engine via Gradio chat with classify, summarize, search, translate, stats, topics, and help intents.
-  **Web Frontend:** Build a production Next.js dashboard deployed to Vercel, providing an interactive demonstration platform beyond Colab/Gradio.

## Midterm Foundation Recap

The midterm NewsBot (Modules 1-8) achieved the following benchmarks that form the baseline for this final project:

| Metric                                                                                                     | Value                                            |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|  <b>Dataset</b>         | 2,000 BBC articles (sampled from 2,225)          |
|  <b>Categories</b>      | Business, Entertainment, Politics, Sport, Tech   |
|  <b>Best Classifier</b> | Logistic Regression — 97.6% accuracy             |
|  <b>Runner-up</b>       | Linear SVC — 97.6% accuracy                      |
|  <b>NER Entities</b>    | 25,637 extracted (PERSON, ORG, GPE, DATE, MONEY) |
|  <b>Highest F1</b>      | Sport — 0.991                                    |
|  <b>Lowest F1</b>       | Politics — 0.957                                 |
|  <b>Live Demo</b>       | Gradio public URL (Colab-hosted)                 |



## 2. System Architecture



### Pipeline Overview

The 2.0 system maintains the original 8-module pipeline as its foundational layer and adds four new modules that consume the pipeline outputs, enriching each article with topic assignments, summaries, multilingual metadata, and conversational accessibility.

| LAYER                                                                                                    | MODULES                                                            | TECHNIQUES                                                      | OUTPUT                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------|
|  Foundation (M1-M8)     | Preprocessing, TF-IDF, POS, Syntax, Sentiment, Classification, NER | spaCy, NLTK, VADER, LogReg / NB / LinearSVC                     | 97.6% classification<br>25,637 NER entities  |
|  M9 — Advanced Analysis | LDA Topic Modeling, Enhanced Sentiment                             | sklearn LDA (10 topics), pyLDAvis, Multi-dim VADER              | Topic distribution<br>Per-category themes    |
|  M10 — Generation      | Abstractive Summary, Semantic Search                               | DistilBART (HuggingFace), SentenceTransformers all-MiniLM-L6-v2 | ~60% compression<br>Cosine similarity search |
|  M11 — Multilingual   | Language Detection, Translation, Cross-lang Classify               | langdetect, deep-translator (Google)                            | 5+ languages<br>auto-classify pipeline       |
|  M12 — Conversational | NL Query Engine, Gradio Chat UI                                    | 7-intent parser, Gradio Blocks API                              | Live chat interface<br>Public shareable URL  |
|  Bonus — Web Frontend | Next.js Dashboard, Vercel Deployment                               | React 18, Tailwind CSS, API routes, Vercel Edge                 | Production web app<br>vercel.app URL         |



### Technology Stack

| Category                                                                                            | Libraries / Tools                    | Version          |
|-----------------------------------------------------------------------------------------------------|--------------------------------------|------------------|
|  Core NLP        | nlk, spaCy, scikit-learn             | 3.8+, 3.8+, 1.4+ |
|  Deep Learning   | transformers, torch, sentencepiece   | 4.38+            |
|  Semantic Search | sentence-transformers                | 2.6+             |
|  Topic Modeling  | gensim, pyLDAvis                     | 4.3+, 3.4+       |
|  Multilingual    | langdetect, deep-translator          | 1.0.9+, 1.11+    |
|  Interface       | gradio                               | 4.0+             |
|  Web Frontend    | Next.js 14, React 18, Tailwind CSS   | LTS              |
|  Deployment      | Vercel, Google Colab                 | Current          |
|  Data            | BBC News Archive (Kaggle, kagglehub) | 2,225 articles   |



3.

## MODULE 9 – ADVANCED CONTENT ANALYSIS



### 3.1 LDA TOPIC MODELING

Latent Dirichlet Allocation (LDA) was applied to the full 2,000-article corpus using a CountVectorizer with 5,000 features (min\_df=5, max\_df=0.9). The model extracted 10 latent topics using online learning with 20 iterations, producing a stable topic structure that aligns meaningfully with the five known categories.



#### CONFIGURATION

- n\_components = 10 topics
- learning\_method = online (mini-batch VB)
- max\_iter = 20
- random\_state = 42 (reproducibility)
- Vocabulary: 5,000 most informative terms
- min\_df = 5 (reduces noise from rare terms)
- max\_df = 0.9 (removes category-agnostic stopwords)



### KEY FINDINGS

Topic coherence revealed strong alignment between the unsupervised LDA topics and the supervised category labels. Sport-related topics were the most semantically concentrated, consistent with the category's high classification F1 of 0.991. Politics topics showed the most overlap — particularly with Business topics — mirroring the classifier's difficulty distinguishing economic-policy articles.

#### REPRESENTATIVE TOPICS BY DOMAIN:



##### Sport:

game | player | team |  
match | season | club |  
win | goal



##### Tech:

technology | software |  
mobile | computer |  
internet | data | digital



##### Business:

market | company |  
share | profit | growth |  
economy | bank



##### Politics:

government | minister |  
party | election | policy |  
vote | parliament



##### Entertainment:

film | music | award |  
star | show | album |  
actor | director



### 3.2 ENHANCED MULTI-DIMENSION SENTIMENT

The midterm VADER sentiment was extended from a single compound score to a full four-dimension profile (compound, positive, negative, neutral) per article. This enables granular analysis of emotional valence distribution across categories and provides richer input features for downstream applications.

#### SENTIMENT PROFILE BY CATEGORY:

| Category      | Mean Compound | Mean Positive | Mean Negative | Label Distribution         |
|---------------|---------------|---------------|---------------|----------------------------|
| Business      | +0.15         | 0.071         | 0.038         | 62% pos / 8% neg / 30% neu |
| Entertainment | +0.53         | 0.132         | 0.024         | 84% pos / 3% neg / 13% neu |

| Category | Mean Compound | Mean Positive | Mean Negative | Label Distribution          |
|----------|---------------|---------------|---------------|-----------------------------|
| Politics | +0.07         | 0.055         | 0.048         | 51% pos / 18% neg / 31% neu |
| Sport    | +0.22         | 0.089         | 0.031         | 69% pos / 6% neg / 25% neu  |
| Tech     | +0.18         | 0.074         | 0.033         | 64% pos / 7% neg / 29% neu  |



4.

## MODULE 10 – LANGUAGE UNDERSTANDING & GENERATION



### 4.1 ABSTRACTIVE SUMMARIZATION

Abstractive summarization was implemented using DistilBART (sshleifer/distilbart-cnn-12-6), a distilled version of BART fine-tuned on the CNN/DailyMail news summarization dataset. DistilBART was selected over full BART for CPU feasibility on Colab free tier while maintaining competitive ROUGE scores on news-domain text.

#### MODEL SELECTION RATIONALE

| Model                         | Params | ROUGE-1 | CPU Speed | Chosen |
|-------------------------------|--------|---------|-----------|--------|
| facebook/bart-large-cnn       | 400M   | 44.16   | Slow      |        |
| sshleifer/distilbart-cnn-12-6 | 306M   | 42.76   | Moderate  | ✓      |
| t5-small                      | 60M    | 38.2    | Fast      |        |
| facebook/bart-base            | 139M   | 40.1    | Fast      |        |



#### SUMMARIZATION PERFORMANCE

On the 5-article category sample, DistilBART achieved an average compression ratio of ~60%, reducing typical 350-word BBC articles to ~140-word summaries while preserving the central facts, named entities, and key sentiment signals. Input is safely truncated to 512 tokens to respect model context limits.



### 4.2 SEMANTIC SEARCH

Semantic search was implemented using the all-MiniLM-L6-v2 sentence transformer, which projects each article into a 384-dimensional embedding space. A 500-article index was built at runtime and queried using cosine similarity via the sentence-transformers util.semantic\_search function.

#### WHY SEMANTIC OVER KEYWORD SEARCH?

Traditional TF-IDF keyword search requires exact vocabulary matches. Semantic search captures meaning: a query for "Premier League transfers" surfaces articles discussing "football player contracts" even with no term overlap. This is critical for the conversational interface where user queries use natural language rather than keyword syntax.



#### SEMANTIC SEARCH CONFIGURATION

- **Embedding model:** all-MiniLM-L6-v2 (22M params, 384 dims)
- **Index size:** 500 articles (RAM-efficient for Colab)
- **Article encoding:** first 256 tokens per article for speed
- **Retrieval:** cosine similarity, top-k configurable (default k=5)
- **Query encoding:** full query text, no truncation
- **Latency:** ~50ms per query on CPU after index is built



## 5. MODULE 11 — MULTILINGUAL INTELLIGENCE



### 5.1 LANGUAGE DETECTION

Language detection was applied across a 400-article sample of the BBC corpus using langdetect, a Python port of Google's language-detection library. The BBC corpus is predominantly English (97%+), with occasional articles in Welsh (cy) and auto-detected noise in very short articles. Seed 42 was set for DetectorFactory to ensure reproducible results.



**Sample Size**  
400 articles



**Primary Language**  
English (97%+)



**Other Languages Detected**  
Welsh (cy), others (<1%)



**Reproducibility**  
Seed 42  
(DetectorFactory)



### 5.2 TRANSLATION PIPELINE

The deep-translator library (Google Translate backend) was integrated to enable automatic classification and sentiment analysis of non-English news articles. The pipeline follows: detect language → translate to English → preprocess → classify → sentiment score. This enables NewsBot 2.0 to function as a language-agnostic news intelligence platform.

#### TRANSLATION DEMO RESULTS

| Source Lang                                                                                         | Article Topic (Original)        | Predicted Category (Post-Translation)                                                        | Sentiment Score (Compound) |
|-----------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------|----------------------------|
|  Spanish (es)    | Medidas económicas del gobierno |  Politics | +0.12                      |
|  French (fr)     | Victoire de l'équipe nationale  |  Sport    | +0.61                      |
|  German (de)     | Quantencomputer-Ankündigung     |  Tech     | +0.38                      |
|  Portuguese (pt) | Cúpula do clima reúne líderes   |  Politics | +0.22                      |
|  Chinese (zh-CN) | AI 助力劳动力市场影响                    |  Tech     | +0.09                      |



All five test articles were correctly routed to their expected categories post-translation, demonstrating the robustness of the Logistic Regression classifier when applied to machine-translated text.



## 6. MODULE 12 — CONVERSATIONAL INTERFACE



### 6.1 INTENT ARCHITECTURE

The conversational engine uses a rule-based intent parser built on keyword matching against a predefined intent taxonomy. This approach was chosen over a fine-tuned NLU model because the command vocabulary is narrow and well-defined, the system needs to run on Colab CPU, and rule-based parsing provides transparent and debuggable behavior appropriate for an academic demonstration.

| Intent                                                                                               | Trigger Keywords                 | Pipeline Steps                     | Output                             |
|------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------|------------------------------------|
|  <b>classify</b>    | classify, predict, what category | preprocess → TF-IDF → LR → VADER   | Category + confidence + sentiment  |
|  <b>summarize</b>   | summarize, tldr, summary         | token-truncate → DistilBART        | Abstract summary + compression %   |
|  <b>search</b>      | search, find, related to         | encode query → cosine similarity   | Top-k ranked article snippets      |
|  <b>translate</b> | translate, foreign, non-english  | detect lang → translate → classify | Translation + category + sentiment |
|  <b>stats</b>     | stats, accuracy, count, how many | stats_cache lookup                 | System metrics dashboard           |
|  <b>topics</b>    | topic, keyword, theme            | stats_cache lookup                 | Top-k keywords per category        |
|  <b>help</b>      | help, commands, guide            | static response                    | Full command reference             |



### 6.2 GRADIO UI IMPLEMENTATION

The Gradio Blocks API was used to build a two-panel chat interface: a primary chat panel with history, quick-example clickable prompts, and a live command reference sidebar. The interface launches with `share=True` to generate a public URL for remote demonstration without requiring local port access.

|                                                                                     |                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|  | gr.Chatbot with bubble_full_width=False for clean message rendering |
|  | gr.Examples for one-click command demonstration                     |
|  | Soft theme with blue primary hue for professional appearance        |
|  | Both button-click and Enter-submit event handlers                   |
|  | Stats sidebar with key system metrics always visible                |

## 7. BONUS: WEB APPLICATION FRONTEND



A production-grade Next.js 14 web application was developed and deployed to Vercel as the 30-point bonus deliverable. The dashboard provides a visual, interactive interface to all NewsBot 2.0 capabilities, targeting non-technical stakeholders such as journalists, editors, and newsroom managers.



### FRONTEND ARCHITECTURE



**Framework:** Next.js 14 with App Router



**Styling:** Tailwind CSS — SST brand palette (gold #C9A84C, black #0B0B0B)



**Charts:** Recharts for classification confidence and sentiment visualizations



**API Layer:** Next.js API routes serving mock pipeline responses (production-ready pattern)



**Deployment:** Vercel with automatic GitHub CI/CD



**Pages:** Dashboard, Classify, Summarize, Search, Translate, About



### KEY DASHBOARD FEATURES



#### Article Classifier

Real-time category prediction with confidence bar chart per category.



#### Summarizer

Before/after word count and compression percentage display.



#### Semantic Search

Top-k article results with relevance score visualization.



#### Translation Panel

Source language auto-detection and machine translation output.



#### System Metrics

97.6% accuracy badge and category breakdown with charts.



#### Responsive Design

Mobile-first layout for tablet/phone use in the field.



## 8.

# PERFORMANCE BENCHMARKS



### CLASSIFICATION PERFORMANCE

| Category      | Precision    | Recall       | F1-Score     | Support    |
|---------------|--------------|--------------|--------------|------------|
| Business      | 0.971        | 0.968        | 0.970        | 80         |
| Entertainment | 0.983        | 0.983        | 0.983        | 60         |
| Politics      | 0.957        | 0.957        | 0.957        | 70         |
| Sport         | 0.991        | 0.991        | 0.991        | 110        |
| Tech          | 0.978        | 0.978        | 0.978        | 80         |
| Accuracy      | —            | —            | <b>0.976</b> | <b>400</b> |
| Macro avg     | <b>0.976</b> | <b>0.975</b> | <b>0.976</b> | <b>400</b> |

Five-category logistic regression classifier on TF-IDF features. Overall accuracy: **97.6%**.



### MODULE RUNTIME BENCHMARKS (GOOGLE COLAB CPU)

| Module               | Operation                    | Approx. Time |
|----------------------|------------------------------|--------------|
| M9 — LDA             | Fit 10 topics, 2,000 docs    | ~45 seconds  |
| M9 — Sentiment       | Score 2,000 articles (VADER) | ~6 seconds   |
| M10 — Summarization  | DistilBART, 5 articles       | ~90 seconds  |
| M10 — Semantic Index | Encode 500 articles (MiniLM) | ~35 seconds  |
| M10 — Search         | Single query retrieval       | < 200ms      |
| M11 — Lang Detect    | 400 articles                 | ~12 seconds  |
| M11 — Translation    | 5 articles (Google API)      | ~15 seconds  |
| M12 — Gradio Launch  | Full interface startup       | ~5 seconds   |



All benchmarks recorded on Google Colab CPU (Free Tier) unless otherwise noted.





## 9. DEPENDENCIES & DEPLOYMENT



### requirements.txt

```
nlk>=3.8
spacy>=3.8
scikit-learn>=1.4
pandas>=2.0
numpy>=1.26
matplotlib>=3.8
seaborn>=0.13
gradio>=4.0
kagglehub>=0.2
transformers>=4.38
torch>=2.0
sentencepiece>=0.1.99
sentence-transformers>=2.6
langdetect>=1.0.9
deep-translator>=1.11
gensim>=4.3
pyLDAvis>=3.4
```



### RUNNING ON GOOGLE COLAB

- 1 Open the notebook via the Colab badge in the GitHub README
- 2 Runtime → Run all (expect ~5 minutes for full pipeline on T4 GPU)
- 3 The Gradio demo auto-launches with a public URL at the final cell
- 4 No API keys required — BBC dataset downloads via kagglehub automatically



### GITHUB REPOSITORY STRUCTURE

```
NewsBot-Intelligence-Engine/
├── ITAI2373-NewsBot-midterm/
├── NewsBot_Intelligence_Engine.ipynb ← M1-M8 midterm
├── ITAI2373-NewsBot-final/
├── NewsBot_Intelligence_Engine_2.0.ipynb ← M9-M12 + bonus
├── newsbot-dashboard/ ← Next.js web app
├── requirements.txt
└── README.md
```



All dependencies are pip-installable in Google Colab (Free Tier).  
The project is configured for zero-secret, zero-key execution.



## 10. CONCLUSIONS & FUTURE WORK



### CONCLUSIONS

NewsBot 2.0 successfully extends the midterm pipeline into a comprehensive, production-adjacent NLP system. The four new modules demonstrate a clear progression from pattern analysis (M9) to language generation (M10) to cross-lingual understanding (M11) to conversational accessibility (M12). The 97.6% baseline accuracy was preserved through the 2.0 rebuild, confirming architectural stability.

The LDA topic model surfaced semantically meaningful themes consistent with human domain knowledge. DistilBART delivered usable summaries at CPU-friendly speeds. The multilingual pipeline correctly classified foreign-language articles across five language families. The conversational interface reduced the system's technical barrier to zero — any user can query the intelligence engine through plain English.



### FUTURE WORK



**Fine-tuned Classification:** Replace Logistic Regression with a domain-adapted DistilBERT for contextual classification, targeting Politics F1 improvement from 0.957 to 0.97+.



**BERTopic Integration:** Replace LDA with BERTopic for neural topic modeling, leveraging SBERT embeddings for more coherent and interpretable topic clusters.



**Real-time RSS Pipeline:** Add live RSS feed ingestion with timestamp tracking to evolve NewsBot from static dataset analysis to streaming intelligence.



**Entity Relationship Graph:** Build a NetworkX entity co-occurrence graph to surface political networks, corporate relationships, and cross-article entity connections.



**FastAPI Productionization:** Wrap the pipeline in a FastAPI REST service with proper authentication for newsroom CMS integration via REST endpoints.



**SST Enterprise Deployment:** Package NewsBot 2.0 as a Southern Shade Technologies product offering for federal agencies requiring structured intelligence extraction from unstructured news corpora.